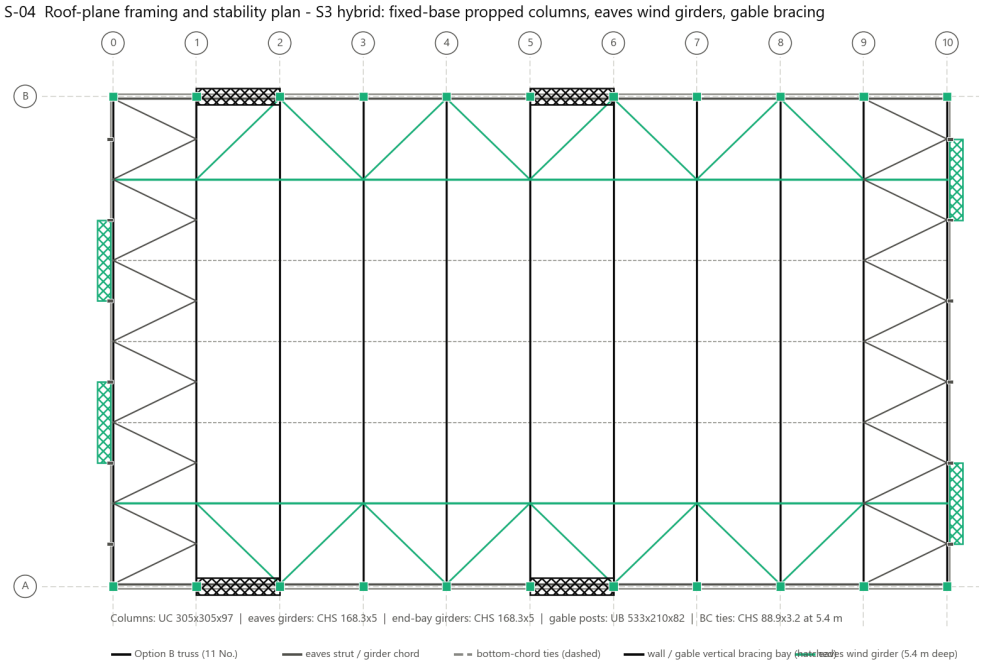
SOUTHGATE WORSHIP CENTER

PRIMARY STEEL SUPPORT STRUCTURE, LATERAL STABILITY & FOUNDATION OPTIMISATION STUDY

Based on the Recommended Option B Roof Truss System

Project	Southgate Worship Center, ~2,000-seat main auditorium, Harare, Zimbabwe
Roof basis	Roof Structure Options Appraisal R2 (7 Nov 2026): Option B, 12-bay bolted paired-angle Warren truss with verticals (fixed)
Model basis	`southgate-bim/bim/southgate.ifc` (IFC4), sha256 6cb86f6d…b035e, read-only
Revision / date	S01 · 7 November 2026
Status	Recommendation for review. Nothing has been implemented in the BIM.



SOUTHGATE WORSHIP CENTER | Option B support-structure study | S-04 | Roof framing S3 | NTS

PRELIMINARY CONCEPT - NOT FOR CONSTRUCTION

This study was prepared with AI assistance under the direction of the client's design team. It contains preliminary analyses and estimates only. It has not been checked, approved or certified by a registered professional engineer (ECZ) and must not be used for construction, fabrication or statutory submission.

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Deliverables in `southgate-bim/studies/option-b-support-foundations/`: A this report; B `data/B_structural_support_register.csv`, `B_footing_register.csv`; C `data/C1–C4_*.csv`, `calcs/*.json`, `scripts/`; D `data/D_steel_framing_comparison.csv`; E `data/E_foundation_comparison.csv`; F `data/F_cost_comparison_model.xlsx`; G `drawings/G_structural_concept_drawings.pdf`; H `data/H_risk_and_outstanding_information_register.csv`; I `review/I_independent_review_record.md`; J `bim_spec/J_BIM_implementation_specification.md`.

1. Executive summary

Recommendation (conditional). For the ground conditions assumed as the base case (G1/G2: allowable bearing ≥ 150 kPa, no expansive clay, no shallow water), support the Option B trusses on the **hybrid stability system S3**: 22 **UC 305×305×97** side columns with **fixed bases about the strong axis**, propped at the eaves by two 5.4 m-deep roof-plane wind girders that span between braced gables; gable wind posts with designed minor-axis restraint; end-bay roof girders and bottom-chord ties; and longitudinal wall bracing in bays 1 and 5. Found the columns on reinforced pads **cast integral with a continuous reinforced strip under the masonry walls (F2)**. Estimated supporting-system cost: **USD 421k (P50)**, P10–P90 389k–457k; deterministic central estimate 399k. **The margin over the braced system S1 is small**: 2.0% at central rates, and S3 is cheaper than S1 in 67% of trials and cheaper than S2 in 89%. **If the site investigation finds poor or wet ground (G3), the braced system S1 with pinned bases becomes the cheaper and simpler choice** (USD 452k against 456k). The stability system should therefore be confirmed after the site investigation (gate G-B).

Answers to the ten questions

1. **Is the existing side-column arrangement suitable?** No. With a complete bracing system, the modelled UC 254×254×73 columns on pinned 4 × M24 bases deflect ≈80–100 mm under the central wind (q_p 0.85 kPa). The limit for the 12.23 m masonry is h/300 = 40.8 mm. The section also fails strength (utilisation ≈1.1) at 1.20 kPa. In addition, two east-gable columns have no footing, there are no gable wind posts, the roof bracing is four 20 mm rods, and the north wall over the loggia has no transfer structure. The grid itself (11 lines @ 5.56 m, columns within the 420 mm wall/pier zone) is suitable and is retained.
2. **Sections and restraint.** S3: UC 305×305×97 (regional stock, by study judgement; 308 mm deep), fixed base about the strong axis, propped at the eaves. Utilisation 0.609; 0.931 with no masonry restraint over the full 13.08 m. Transverse deflection 26.1 mm and longitudinal eaves sway 40.7 mm against 40.8 mm. S1 needs UC 305×305×137 (pinned) and S2 needs UC 356×368×153 (imported, by study judgement). The UB alternatives no longer pass once the purlins are not relied on as girder chords, and in any case they depend on masonry restraint.
3. **Braced, moment or hybrid?** Hybrid on firm ground, braced on poor ground. S2 is the most expensive in every soil scenario (5% above S3 at G2) and is not recommended. The gap between S3 and S1 (8k) is within the estimating uncertainty, so non-cost factors also count. S3 is self-stable during erection, has transverse stiffness reserve and uses the lightest columns. S1 has simpler pinned bases and smaller pads, which win where the ground is poor.
4. **Can the gallery or integrated framing help?** Not in the recommended scheme. The gallery reaches only +7.85 m, has no designed foundations and carries placeholder sections. The stair cores (walls to +10.80 m, construction unverified, no footings) could replace the east-gable bracing and the bay-5 wall bracing if they are proven to be RC shear walls. That is a conditional optimisation estimated at USD 10–15k, not a basis for design. The single-storey office wing should remain structurally independent.
5. **Most economical credible foundation.** F2, reinforced pads cast integral with a 0.60 × 0.45 m reinforced wall strip, is the cheapest arrangement for every system in every soil scenario. The wall weight is counted only where the wall actually exists: net of doors, the north loggia and the east glazed slot. For S3 the pads are 2.3–3.3 m square and 0.80 m deep (the depth is set by the holding-down anchorage). Under G3, F3 (deep pads with ground beams over void formers) costs only about USD 1,136 more and is the prudent choice if the clay is expansive.
6. **Governing foundation actions.** Uplift and overturning under the concurrent ULS combinations govern, not bearing. Braced-bay columns see ≈309 kN uplift and the braced gable posts ≈420 kN. The S3 fixed bases carry ≈200–220 kNm together with up to ≈290 kN of uplift. Ordinary truss bearings deliver 55–92 kN uplift (q_p 0.85–1.20). The construction stage, before the masonry is built, governs several pads.
7. **Quantities (S1 / S2 / S3).** Supporting steel 84 / 79 / 74 t. Foundation concrete 190 / 182 / 229 m^3. Reinforcement, including anchor reinforcement, 6.1 / 8.1 / 8.9 t. Excavation 649 / 626 / 746 m^3 (§13).
8. **Costs (central, G2/F2).** USD 406,705 / 420,305 / 398,792. P50 429k / 441k / 421k. VAT and the common Option B roof package (≈USD 195k, R2) are excluded. S3 has the least steel and the most concrete.
9. **Most robust.** On firm ground, S3: it keeps its stiffness reserve transversely and is self-stable in erection. Selected-design sensitivity failures: S1 SLS wind at 1.20 kPa, Dominant opening cpi +0.55, Dominant opening in suction zone, cpi -1.08 (factored 1.3); S2 SLS wind at 1.20 kPa, Base fixity: rotational spring 50 MNm/rad (G2), Base fixity: rotational spring 20 MNm/rad (G3); S3 SLS wind at 1.20 kPa (§15). On poor or wet ground, S1. **Two assumptions**

can reverse the choice: the ground (G3 favours S1) and the wall construction (lightweight cladding removes the masonry stiffness criterion; §15).

10. **Information needed before detailed design.** Confirmation of the wall construction; the Harare design wind speed and the governing code set; a site investigation (trial pits, DPSH/boreholes, swell and collapse tests, groundwater, subgrade modulus for base fixity); equipment and rigging weights; supplier prices for UC 305, CHS and holding-down assemblies; and the construction sequence (§18).

> **Option B issues reported, not redesigned.** (a) When the gable trusses act as lateral-system collectors, their 2L80×8 top chords reach axial utilisation ≈0.8–0.9 before purlin bending, against 0.72 in R2. Consider 2L90×8, as on the interior lines. (b) The gable bottom-chord heel panels need out-of-plane restraint under uplift (elastic critical factor 0.22 without it). The eigen-analysis also shows the gable bottom chord between ties at $\alpha \approx 2.0$, which is a truss-member check for the Option B design. (c) The truss bearing as modelled is 145 mm eccentric to the column axis. (d) The 3D truss reactions are 10–25% above R2's 2D screening values (Table 3.2). (e) The bearing bolts act as ties in wind and should be preloaded (slip-resistant) or fitted, not ordinary bolts in slotted holes.

2. Project description and study scope

The auditorium is a 55.9 × 33.0 m hall. Eleven Option B trusses span 32.4 m between side-wall columns at 5.56 m centres (grid lines 0–10, x = 3.25–58.85 m). The stage is at the west end (line 0), and the gallery and stair cores are at the east (x 36.6–58.7 m). A single-storey office wing (+4.42 m) adjoins the west gable. The side walls are 300 mm masonry infill between the columns, 12.5 m high, with clerestory windows at +10.10 to +11.80 m in bays 0–6 and doors in bays 0, 4, 7 and 9. A 9.5 m loggia opening on the north side (x 43–52.5 m) rises to +4.00 m.

Scope: roof support columns, side-wall and gable framing, longitudinal and transverse stability, roof-plane girders and bracing, bottom-chord restraint, truss-to-column and base connections, foundations and ground beams, and the interfaces with the gallery, entrances, cores and office wing. **Excluded:** the detached children's church, other standalone buildings, the Option B truss configuration (fixed), and the gallery, core and office structures (interfaces only). The study did not modify `southgate.ifc` or the original design project. All analysis used isolated study models.

2.1 Method

1. Read-only audit of the authoritative IFC (sha256 verified against `bim/publish_record.json`) and the migration audit reports.
2. An independent 3D frame model (`scripts/model_b.py`, `frame3d.py`) of all 11 Option B trusses (R2 member sizes), 22 columns, 12 gable posts and every bracing system. Solver verified against closed-form results (Appendix A).
3. Characteristic load cases with SANS 10160-1 combinations. Eight wind cases (four directions, two internal pressures) at q_p 0.85 and 1.20 kPa.
4. For each stability system, a sweep over 6–9 column sections. In each run the girders, bracing and posts are re-sized for strength and then stiffened until the SLS limits are met.
5. Pad design for every base: three soil scenarios × three arrangements. Bearing, uplift, overturning, sliding, bending, shear and punching are checked, and anchor embedment is coupled to pad depth.
6. Line-item costing (quantity × rate × factor), Monte Carlo on correlated rate groups, and sensitivity re-runs of the sizing.

3. Recommended roof-truss design basis

Option B is the fixed basis: a duo-pitch Warren truss with verticals, 12 panels @ 2.70 m, span 32.40 m (bearings at y = 0.30 / 32.70 m), 15° pitch, top chord +12.38 m at the heel and +16.72 m at the ridge, bottom chord +10.90 m, heel depth 1.48 m, ridge depth 5.82 m, 11 trusses @ 5.56 m, ≈42.1 t. The geometry was verified against the IFC truss assemblies (bounding boxes x = 3.12…58.98, z = 10.67…16.77 m). Member sizes follow the R2 central run: TC 2L90×8 (gables 2L80×8), BC 2L80×8, end diagonals 2L100×8, other diagonals 2L75×6, verticals 2L60/2L75×6, king post 2L100×8, end posts 2L70×6. The truss was not re-optimised.

3.1 Bearing condition

The trusses bear at the top-chord heel on column cap plates at +12.28/+12.38 m. In S1 and S3 both bearings are pinned (bolts finally tightened after the roof dead load is applied), so the roof props both column heads. Option B's gravity spread is only 1–2 mm, so the resulting thrust is small (it is included in the analysis). In S2 the bottom chord is also connected to the column at +10.90 m (a 1.48 m "knee"), which makes each truss line a portal. The R2 bearing seat does not provide moment continuity, so S2 needs a purpose-designed heel connection.

3.2 Support reactions (S3 model, ULS)

q_p kPa	max compression kN	interior uplift kN	gable-line uplift kN	H transverse kN	H longitudinal kN	R2 screening (B)
0.85	123.7	-62.1	-62.3	38.2	37.2	110 / −44
1.20	144.8	-103.3	-101.4	54.0	52.1	– / −75 (gable −104)

The 3D reactions are 10–25% above R2's 2D values. They include the eaves girders' share of gravity, frame action, equipment on lines 1–2 and the near-edge roof zones. The upper-scenario uplift (−92 kN interior, −102 kN at the gables) governs the holding-down design.

4. Existing side steel framing and foundation audit

Source: `scripts/s02_extract_inventory.py` and `s04_register.py` (read-only), giving 149 register rows (deliverable B). Findings were checked against `independent_final_review.md` O-06 and `outstanding_work.csv` OW-001/-028.

Element	IFC evidence	Verified status	Engineering finding
Roof columns (22)	SOUTHGATE-COL-0011…0032, UC 254×254×73.1, z −0.80→+12.28, y 0.155 / 32.845	CONCEPT_UNVERIFIED	Section inadequate for masonry deflection and for strength at 1.20 kPa (§6)
Roof-column footings	20 concept pads 1.5×1.5×0.65 m (lines 1–9), 1.6×1.3×0.65 m (line 0) + west strip; **none at line 10** (SOUTHGATE-COL-0015, SOUTHGATE-COL-0016)	CONCEPT_UNVERIFIED	Too small for uplift (they provide ≈35 kN against 55–310 kN needed); no strip under the long walls
Base plates	SOUTHGATE-PLATE-0576…0597, 450×450×30, 4 M24 (schematic)	concept	Pinned; no anchorage design; cone breakout not checked
Caps / seats	PLATE-0664…0707 at +12.18…+12.56	concept	Bearing 145 mm eccentric to the column axis
Roof bracing	SOUTHGATE-BRACE-0001…0004, 20 mm rods, bays 0 and 9 only	CONCEPT_UNVERIFIED	No eaves girders, cross frames, bottom-chord ties or wall bracing
Gable wind posts	none	–	**Missing.** 32.4 m gable walls have no intermediate vertical support
Gallery columns	10 columns (RECT 400/300 placeholders); 7 rear columns with no footing	CONCEPT_UNVERIFIED	Not engineered; not usable for stability
Stair cores	CORE-* walls 150–200 mm, to +10.80 m, no footings	CONCEPT_UNVERIFIED	Construction unknown; candidate stability elements only if proven
Masonry piers / walls	FACADE-*-BAY-*-BRICK-INFILL (0.30 m, to +12.50) + 260×500 piers	concept; render-material mass	Masonry design intent (source schedule); 72 kN/m self-weight needs a strip footing
North loggia	EXT-N-19…21 walls start at +4.00 over a 9.5 m opening	existing design reference	No lintel / transfer beam modelled

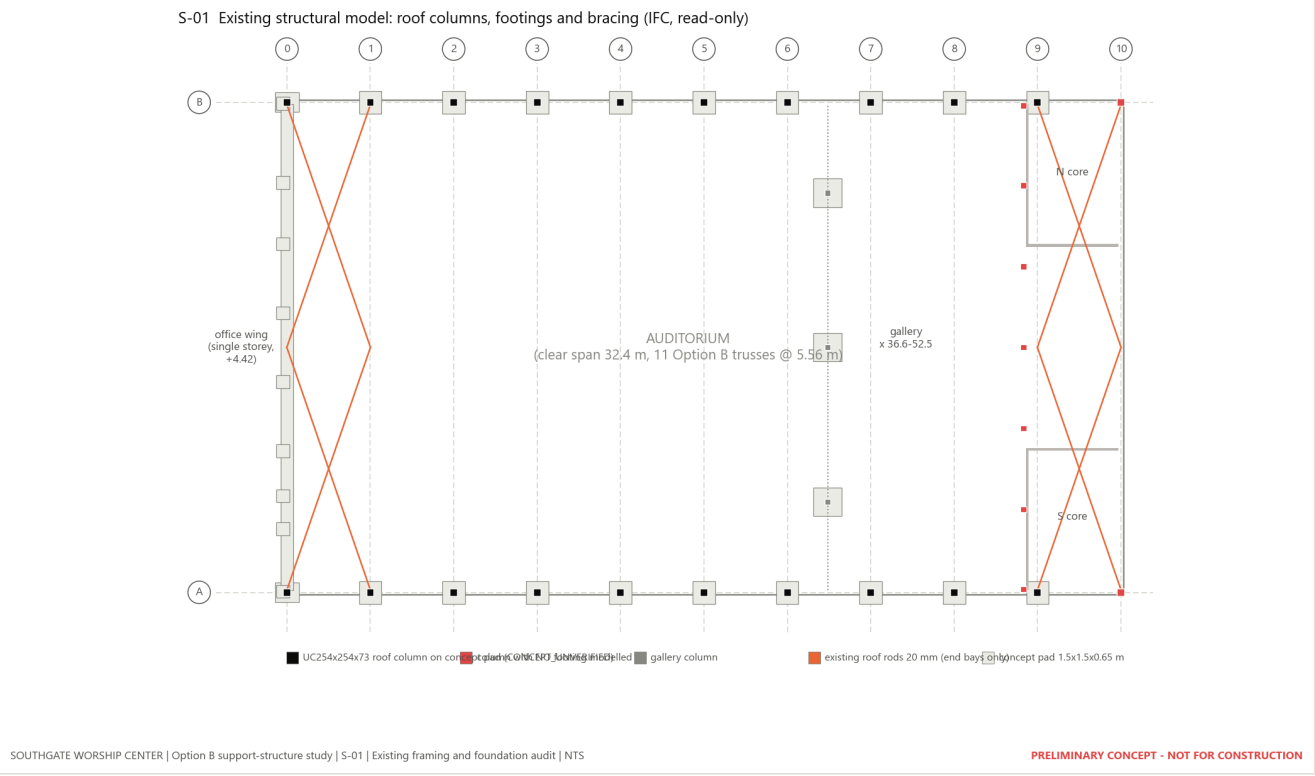


Figure 4.1 Existing structural model: columns (red: no footing), concept pads, cores and roof rods.

Distinction: every structural object in the IFC has a correct class, profile and stable identity, but none is engineered. Classification does not establish adequacy; this study treats all of them as geometry to be verified.

5. Structural loading and design assumptions

5.1 Design standards

No Zimbabwean loading or steel design standard could be confirmed from published sources. Statutory Instrument 41 of 1978 references a Standards Association of Zimbabwe concrete code, and the Model Building By-laws govern approvals, but neither sets wind or steel design rules found in this review. Consistent with R2 and regional practice, the study uses **SANS 10160-1/-2/-3 (actions)**, **SANS 10162-1:2011 (steel, limit states, φ 0.9, n = 1.34)**, **SANS 10100-1 (concrete, BS 8110 basis)** and **EN 1992-4** (anchorage, by analogy), with EN 1991-1-4 informative pressure coefficients. These are suitable for the next design stage *only if* the appointed ECZ-registered engineer and Harare City Council accept them. Edition and consequence class must be confirmed: CC2 is recommended for a 2,000-seat assembly building (reliability class RC2 minimum; consider RC3 for the roof given the occupancy).

5.2 Loads

Action	Value	Basis / status
Roof dead (sheeting, purlins, insulation, bracing allowance)	0.178 kN/m² plan	R2 load schedule (model quantities + typical supplier masses)
Truss / steel self-weight	members × 1.05–1.08	calculated
General suspended services	0.10 kN/m²	R2 central; future 0.25
Acoustic rafts, HVLS fans, ridge fans, trays	3.4 kN/group, 1.8 kN, 4.0 kN, 0.25 kN/m	scenario: no products selected
Rigging (line arrays 8 kN, lighting bars 0.40 kN/m), dynamic ×1.2	imposed, γ 1.6	scenario; future scenario as sensitivity
Roof imposed (inaccessible)	0.25 kN/m²	SANS 10160-2 category H, to verify
Peak velocity pressure q_p	0.85 kPa (central, SLS); 1.20 kPa (upper, ULS and holding-down)	R2 study values (≈41/49 m/s gust); **Harare design gust not confirmed**
Wall c_{pe}: long walls D/E; gables D/E; parallel A/B/C	+0.72/−0.37; +0.70/−0.30; −1.2/−0.8/−0.5	EN 1991-1-4 Table 7.1 (h/d 0.44 / 0.26; e = 29 m)
Internal c_{pi}	+0.2 / −0.3 (dominant opening +0.55 sensitivity)	large doors may open in storms: confirm management
Roof net c_p	R2 duo-pitch profiles, −1.5 at gable lines	EN 1991-1-4 Table 7.4a
Masonry walls	72 kN/m (300 mm, 19 kN/m³, 13.3 m, less openings)	assumption: wall type not confirmed
Loggia wall over the north opening	134 / 267 / 134 kN on lines 7/8/9 north	assumed transfer beam (not modelled)
Notional loads	0.005 × factored gravity	SANS 10162-1 cl 8.7.2
Seismic, snow, ponding	not governing	R2 assessment

5.3 Combinations and criteria

- ULS: 1.35G; 1.2G + 1.6Q_r; 1.2G + 1.6Q_{rig}; 1.2G + 1.3W; 0.9G + 1.3W (equipment excluded where favourable), each with notional loads.
- SLS: 1.0W at the characteristic central pressure q_p 0.85 kPa. Masonry cracking is a serviceability state, so the central basis is used, and 1.20 kPa is a sensitivity case. **Limit h/300 = 40.8 mm**, with h = 12.23 m of masonry (FFL to wall head), for column deflection *and* longitudinal eaves sway, because the masonry infill spans to the columns. R2 used h/150, which suits lightweight cladding only. Posts: L/300 relative to the chord.
- Foundations: allowable bearing under characteristic combinations (Meyerhof effective area); uplift, overturning and sliding under EQU factors (0.9 on permanent stabilising); construction stage with walls absent at 0.75 × W(1.20 kPa) (assumed reduced exposure period).
- Second order: $\alpha_{cr} = (H/V)(h/\delta)$ per frame line and wall line; first-order analysis is acceptable if $\alpha_{cr} \geq 10$. An elastic eigen-analysis was used to find member-level modes.

6. Existing structural system assessment

Checking the modelled arrangement (UC 254×254×73, pinned bases) in the 3D model with a complete S1 bracing system added:

Check	Result	Limit / capacity	Verdict
Column deflection, SLS 0.85 kPa	≈101 mm (sizing start) / 81 mm column bending alone (hand: $5wL^4/384EI$, w = 4.82 kN/m)	40.8 mm (h/300, h 12.23 m)	fails ×2.5
Column strength, ULS 1.20 kPa (zone A suction, line 1)	utilisation 1.09 (M ≈ 205 kNm vs φM_p 317 kNm, plus minor-axis moment)	1.00	fails
Concept pads 1.5×1.5×0.65 m	weight ≈ 35 kN (+ soil ≈ 30 kN) × 0.9	55–92 kN uplift (bearing), 290–310 kN (braced bays)	fails
4 M24 pinned base	adequate in tension for ordinary bearings; cone breakout needs hef ≥ 300 mm	no design	concept only
Roof bracing (4 × 20 mm rods)	capacity ≈ 80 kN each	end-bay girder diagonals 110–130 kN, plus eaves girders	inadequate
Gable walls	no wind posts	32.4 m × 12.5–16.9 m walls need vertical support	missing
Gable BC heels under uplift	eigen α_{cr} 0.22 without heel restraint	≥ 1 (member)	restraint needed

R2 had shown the UC 254 column passing with a roof wind girder (sway ≈80 mm against h/150). The masonry criterion, the zone-A suction on the end columns, and the gable posts are what change the conclusion. R2's lateral system allowance (≈10.6 t of eaves girders, 22 pads for 80 kN) also covered only a small part of the necessary supporting structure: the complete systems here use 74–84 t of supporting steel.

7. Lateral stability alternatives

All three systems share the **longitudinal** system, because longitudinal moment frames would put the columns into minor-axis bending and are impractical. That system comprises: six gable wind posts per gable (UB 533×210×82, placed at s = 2.7, 8.1…29.7 m so as to clear the west doors and the east glazed slot); end-bay roof girders in bays 0 and 9 across the full width; end-bay cross frames and bottom-chord bracing; bottom-chord ties at 5.4 m; eaves struts; and wall bracing in bays 1 and 5 on both sides. The north wall has no bay free of clerestories or openings, so the wall bracing is an X below the sill (+9.90 m) with knee braces inside the 1.08 m window piers.

	S1 Braced frame	S2 Moment-resisting	S3 Hybrid
Transverse system	Eaves wind girders (TC plane, 5.4 m deep) → gable X-bracing	Each truss line a portal: fixed bases + TC/BC heel knee	As S1, plus fixed column bases (propped cantilevers)
Column bases	pinned (4 M24)	fixed, strong axis	fixed, strong axis
Truss bearing	pinned, TC heel	TC + BC knee (moment-capable, to design)	pinned, TC heel
Gable bracing	2 two-tier X panels per gable	none (posts only)	as S1
Erection stability	columns unstable until girders are complete	self-stable cantilevers	self-stable cantilevers
α_{cr} transverse / longitudinal	48.4 / 42.6	61.7 / 73.6	87.3 / 33.5

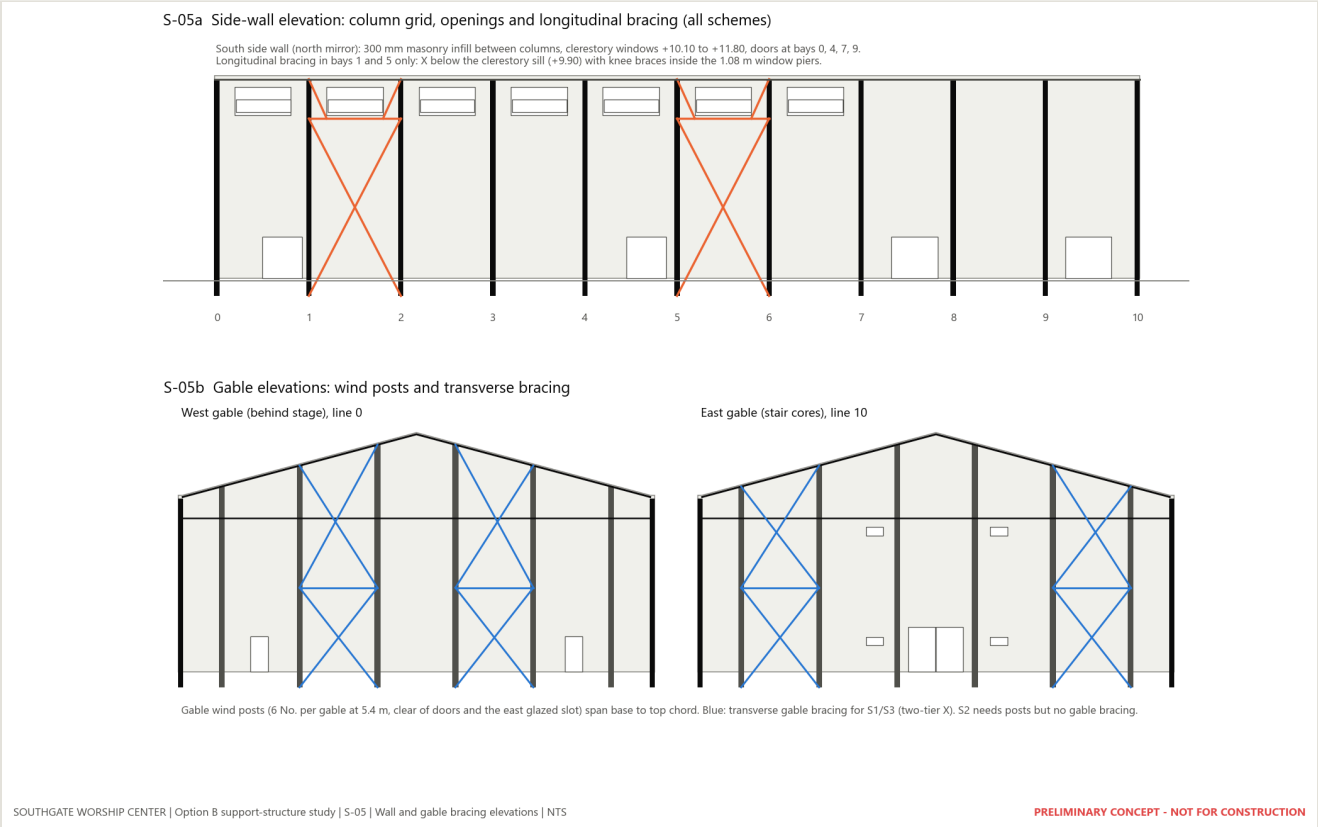


Figure 7.1 Longitudinal wall bracing (bays 1 and 5) and gable wind posts / transverse bracing.

Variants considered and not adopted. (i) *Cantilever columns without girders* (R2): no catalogue UC meets the deflection limit. (ii) *Cores as the east stability elements* (S3-core): this would remove the east-gable and bay-5 bracing, but it depends on unverified RC cores that have no foundations. It is held as a conditional optimisation (§10.4). (iii) *Intermediate moment frames to shorten the girders*: frames stiff enough to help the girders need UC 356 columns, which costs more than the girders they replace. (iv) *Masonry gables as shear walls*: credible in Harare practice, but it needs reinforced masonry design and a steel erection sequence that does not depend on the masonry. Recorded as an optimisation.

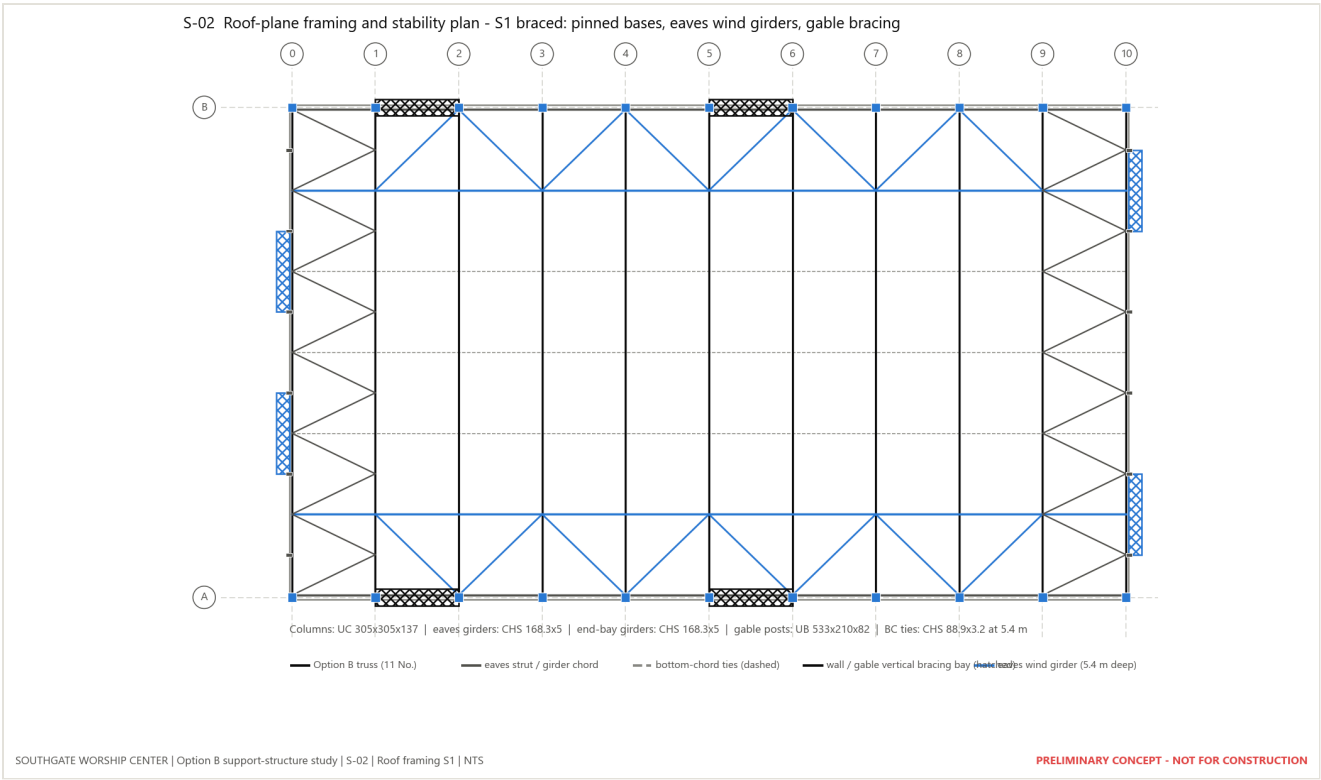


Figure 7.2 S1 roof-plane framing.

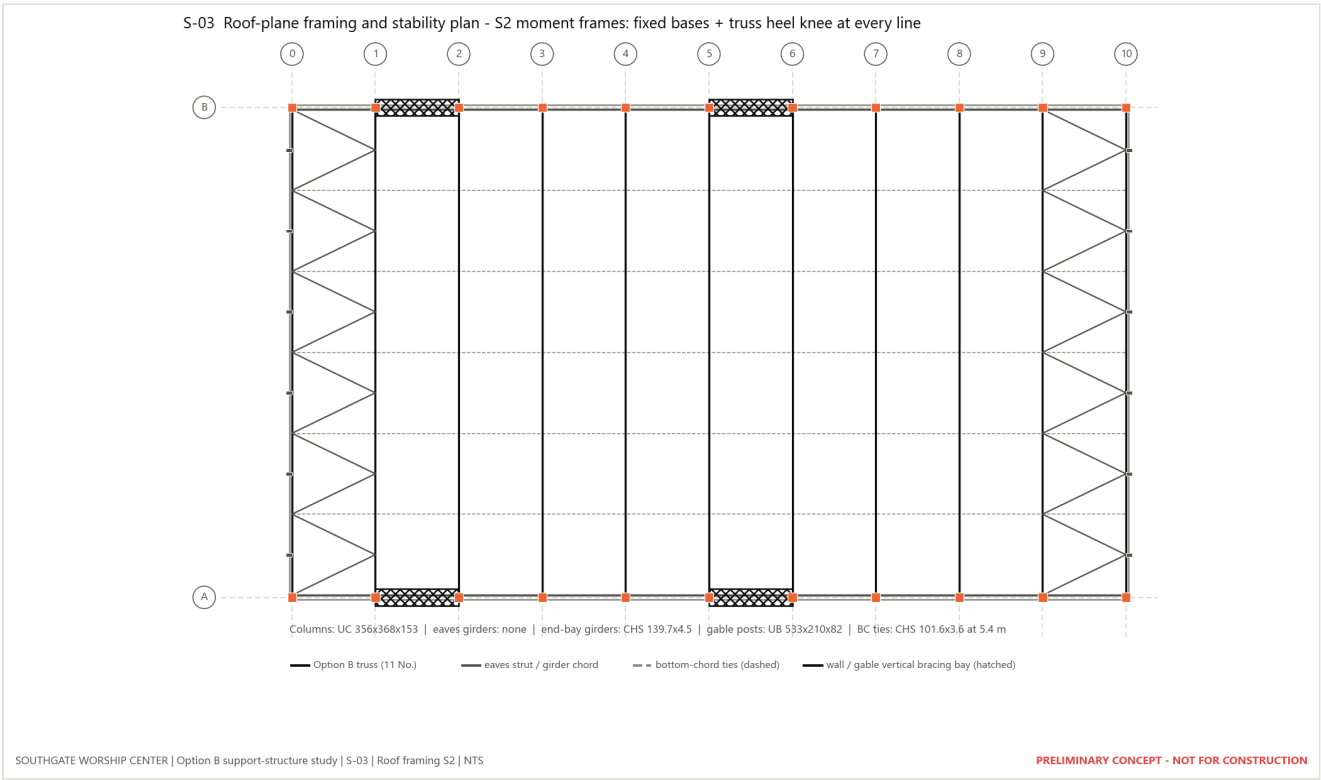


Figure 7.3 S2 roof-plane framing (no eaves girders).

8. Side-column optimisation

Each candidate was analysed with its complete system re-sized (Figure 8.1, Table 8.1). Column checks follow SANS 10162-1 cl 13.8 (cross-section, overall member and lateral-torsional buckling), with K_x = 1.0 (S1, S2) or 0.85 (S3), minor-axis and LTB restraint at 3.3 m from the masonry ties (robustness check: no restraint over the full 13.08 m), ω_2 = 1.13, and the bearing eccentricity moment included. The governing case in all systems is longitudinal wind with zone-A suction on the end bays plus wall-bracing forces in the braced-bay columns.

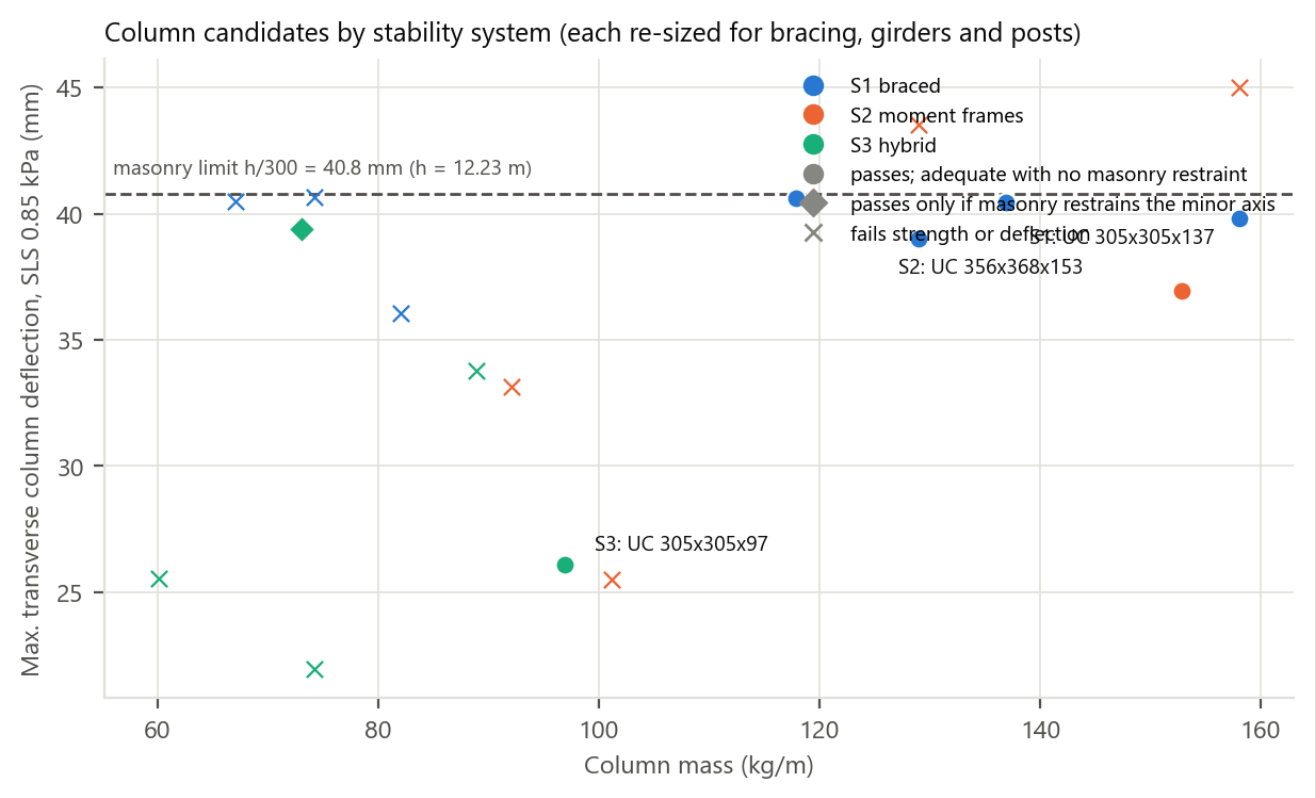


Figure 8.1 Column mass against SLS deflection. Diamonds pass only if the masonry restrains the minor axis.

System	Column	kg/m	depth mm	availability (judgement)	util.	transverse SLS mm	base case (both SLS limits)	util. with no masonry restraint	total USD (G2/F2)
S1	UB 457x191x67	67	453 ⚠	SA-stock	1.12	40.5	fail	30.78 fail	–
S1	UB 406x178x74	74	413	SA-stock	1.01	40.7	fail	59.96 fail	–
S1	UB 457x191x82	82	460 ⚠	SA-stock	0.88	36.1	fail	2.96 fail	–
S1	UC 305x305x118	118	314	SA-stock	0.61	40.6	pass	0.80	431,515
S1	UC 356x368x129	129	356	import	0.48	39.0	pass	0.61	414,954
S1	UC 305x305x137	137	320	SA-order	0.53	40.5	pass	0.64	406,705
S1	UC 305x305x158	158	327	SA-order	0.45	39.8	pass	0.53	415,884
S2	UB 533x210x92	92	533 ⚠	SA-order	0.69	33.1	fail	2.42 fail	–
S2	UB 610x229x101	101	603 ⚠	SA-order	0.67	25.5	fail	2.06 fail	–
S2	UC 356x368x129	129	356	import	0.42	43.5	fail	0.63	–
S2	UC 356x368x153	153	362	import	0.35	36.9	pass	0.49	420,305

System	Column	kg/m	depth mm	availability (judgement)	util.	transverse SLS mm	base case (both SLS limits)	util. with no masonry restraint	total USD (G2/F2)
S2	UC 305x305x158	158	327	SA-order	0.39	45.0	fail	0.56	–
S3	UB 406x178x60	60	406	SA-stock	1.20	25.5	fail	66.54 fail	–
S3	UC 254x254x73	73	254	SA-stock	0.73	39.4	pass	1.65 fail	418,045
S3	UB 406x178x74	74	413	SA-stock	0.94	21.9	fail	45.38 fail	–
S3	UC 254x254x89	89	260	SA-stock	0.73	33.8	fail	1.19 fail	–
S3	UC 305x305x97	97	308	SA-stock	0.61	26.1	pass	0.93	398,792

⚠ depth over 420 mm: projects beyond the wall/pier zone. Not recommended without architectural change.

8.1 Selection logic

A candidate must (a) pass strength and both SLS limits (transverse deflection and longitudinal sway, 40.8 mm), (b) remain adequate with **no masonry restraint at all**, checked over the full 13.08 m column height, and (c) fit the 420 mm wall/pier zone. Candidates deeper than 420 mm were also costed with the pier enlargement priced, and none of them passes (a) and (b). The resulting columns are UC 305×305×137 for S1, UC 356×368×153 for S2 and **UC 305×305×97 for S3**. Fixing the bases makes each column ≈2.4× stiffer against wall wind ($wL^4/185EI$ against $5wL^4/384EI$), which is why S3's column is 40 kg/m lighter than S1's. The existing UC 254×254×73 passes in S3 with masonry restraint (39.4 mm, utilisation 0.73) but fails without it (1.65). It is acceptable only if the masonry ties are designed as restraints.

Method limitation. The stiffening loop upsizes one bracing group at a time, choosing the group with the largest strain energy per tonne under the governing SLS case. It is a heuristic. The S3 UC 254×254×89 result (failing longitudinal sway while the lighter UC 254×73 passes) is a path artefact of the heuristic, not a property of the section. Costs from the sizing method are uncertain by ≈±5–10%, and this is carried into the Monte Carlo (§15).

Gable posts: UB 533×210×82 posts are adequate only with minor-axis restraint. Without it, utilisation is 2.6–4.6 over the 14–17.5 m height. All systems therefore include two rows of horizontal restraint struts per gable (CHS 88.9×3.2, ≈0.9 t), and the masonry ties must not be relied on alone.

Construction stage: S2 and S3 columns are self-stable cantilevers once grouted. S1 columns need guying until the girders and gable bracing are complete (priced as temporary works).

9. Structural bracing and connection concepts

Member group	S1	S2	S3
Side columns (lines 1-9)	UC 305x305x137 · 32.2 t	UC 356x368x153 · 36.0 t	UC 305x305x97 · 22.8 t
Gable-line columns (0, 10)	UC 305x305x137 · 7.2 t	UC 356x368x153 · 8.0 t	UC 305x305x97 · 5.1 t
Gable wind posts (12)	UB 533x210x82 · 15.1 t	UB 533x210x82 · 15.1 t	UB 533x210x82 · 15.1 t
Eaves struts	CHS 168.3x6.3 · 2.8 t	CHS 139.7x4.5 · 1.7 t	CHS 168.3x6.3 · 2.8 t
Eaves girder inner chord	CHS 168.3x5 · 2.2 t	– (not required)	CHS 139.7x4.5 · 1.7 t
Eaves girder diagonals	CHS 168.3x5 · 2.5 t	– (not required)	CHS 168.3x5 · 2.5 t
End-bay roof girder diagonals	CHS 168.3x5 · 3.0 t	CHS 139.7x4.5 · 2.2 t	CHS 168.3x5 · 3.0 t
End-bay cross frames	CHS 139.7x4.5 · 2.1 t	CHS 139.7x4.5 · 2.1 t	CHS 139.7x4.5 · 2.1 t
End-bay bottom-chord bracing	CHS 139.7x4.5 · 0.9 t	CHS 139.7x4.5 · 0.9 t	CHS 139.7x4.5 · 0.9 t
Bottom-chord ties @ 5.4 m	CHS 88.9x3.2 · 1.9 t	CHS 101.6x3.6 · 2.4 t	CHS 88.9x3.2 · 1.9 t
Gable bracing diagonals	CHS 219.1x6.3 · 5.0 t	– (not required)	CHS 193.7x6.3 · 4.4 t
Wall bracing X (bays 1, 5)	CHS 193.7x6.3 · 2.8 t	CHS 193.7x6.3 · 2.8 t	CHS 219.1x8 · 4.0 t
Wall bracing knees	CHS 60.3x3 · 0.1 t	CHS 48.3x3 · 0.1 t	CHS 114.3x5 · 0.3 t
Wall bracing sill strut	CHS 101.6x3.6 · 0.2 t	CHS 101.6x3.6 · 0.2 t	CHS 101.6x3.6 · 0.2 t

All bracing is CHS (S355J2H), designed for compression with both diagonals active, so that the stiffness assumed in the model is the stiffness provided. Tension-only rods would be lighter but would halve the X-panel stiffness, which the deflection limits do not allow without re-analysis. **Purlins are not relied on as girder chords.** In the design runs the purlins are nominal links (0.1% of their axial stiffness), so the eaves struts and inner chords carry all the girder bending. The independent review found that an earlier draft let the purlins share it.

9.1 Connection concepts (not connection designs)

- **Truss bearing (S1/S3):** concentric cap plate on the column (eliminating the modelled 145 mm offset, or designing the column for $R_v \times 0.145 \approx 18–21$ kNm). **Preloaded (slip-resistant) M20 8.8 bolts**, because the bearing acts as a tie for the eaves girders in wind; slotted holes only if the bolts are preloaded after the roof dead load. Heel restraint cleats at gable lines 0 and 10.
- **Knee (S2):** TC heel and BC heel both bolted to the column flange. Knee forces ≈229 kN horizontal at ULS. The paired-angle BC and gussets at the heel need re-design for this force (a truss change outside this study's mandate, and a further reason S2 is not recommended).
- **Bracing ends:** slotted-tube to 10–12 mm gusset, 2–4 M20 8.8 bolts, gusset welded to the column flange or the truss node plate. The wall X-bracing is made in two halves joined at a central node plate (CHS 219 cannot cross inside a 300 mm wall), placed on the inner face or in a wall chase. The lower ends connect to gussets on the column feet below FFL, inside the concrete encasement pocket.
- **Gable posts:** base pinned (4 M24), head connected to the gable truss TC node through a vertically slotted bracket (horizontal restraint, vertical freedom).
- **Bases:** Figure 9.1 and §11.

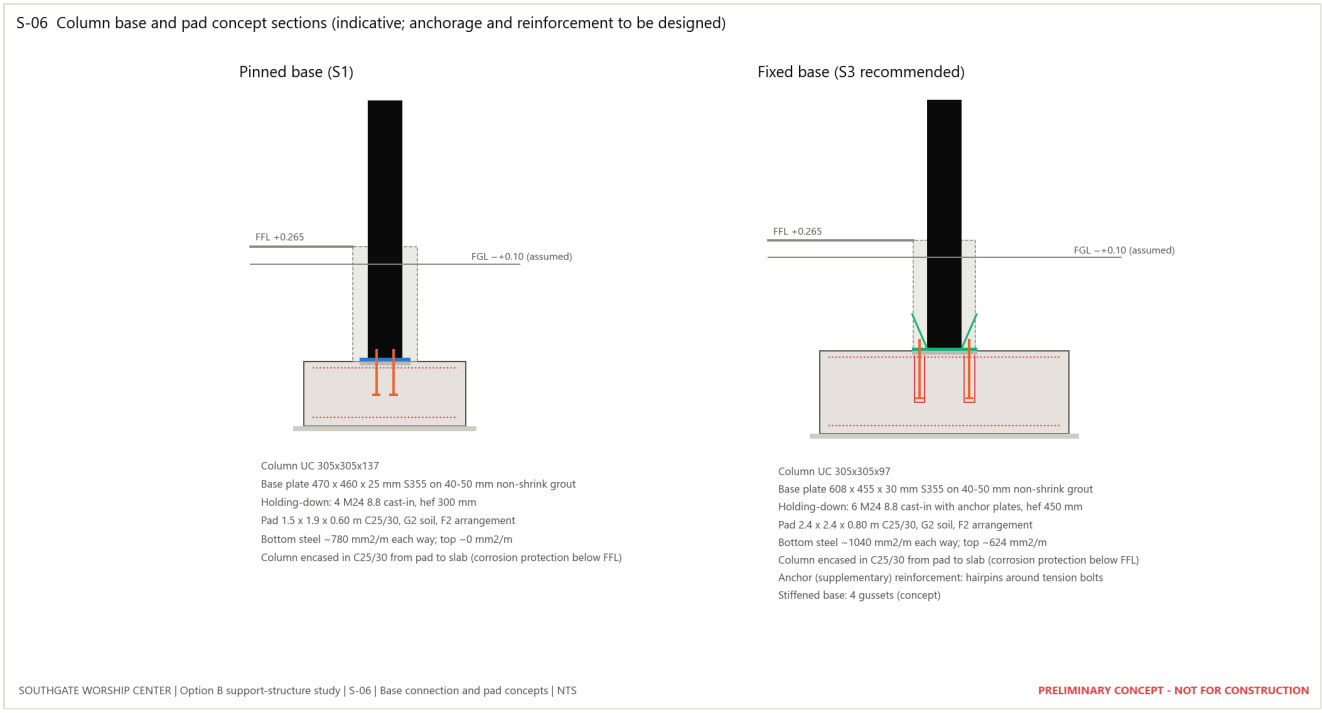


Figure 9.1 Pinned (S1) and fixed (S3) base and pad concept sections, indicative.

10. Foundation optimisation

10.1 Ground conditions

No geotechnical report exists for the site (project-wide search). Published Harare evidence is limited. A University of Zimbabwe study recorded intermediate- and low-plasticity clays, sodic soils and inadequate bearing in places, and recommended a minimum founding depth of 1.5 m for bearing above 100 kPa. Granite-derived residual soils in the region can be collapsible. The study therefore uses three scenarios, all **assumptions**:

Scenario	Allowable bearing	Min. founding below FGL	Density	Base friction	Groundwater / heave
G1 favourable: dense residual granite / gravelly sand	200 kPa	1.0 m	18 kN/m³	0.45	none
G2 moderate (base): stiff residual sandy clay	150 kPa	1.2 m	18 kN/m³	0.35	none
G3 poor: firm intermediate-plasticity clay, possibly expansive	100 kPa	1.5 m	17 kN/m³	0.30	perched water at −1.40 (buoyant uplift); heave → F3

Below about 75 kPa allowable, or with high swell potential, pads and strips stop being appropriate and a raft or piles would be required. That is outside this study and is identified as the decision threshold.

10.2 Arrangements

- **F1** independent pads (no wall weight relied on) plus a lightly reinforced wall strip.
- **F2** pads cast integral with a continuous reinforced strip under the masonry. The wall weight over 3.0 m (216 kN) counts as permanent ballast. Construction stage checked with the walls absent.
- **F3** deeper pads with 0.35 × 0.75 m ground beams spanning 5.56 m that carry the wall (void formers in G3). The full bay of wall load (≈400 kN) goes into the pads.

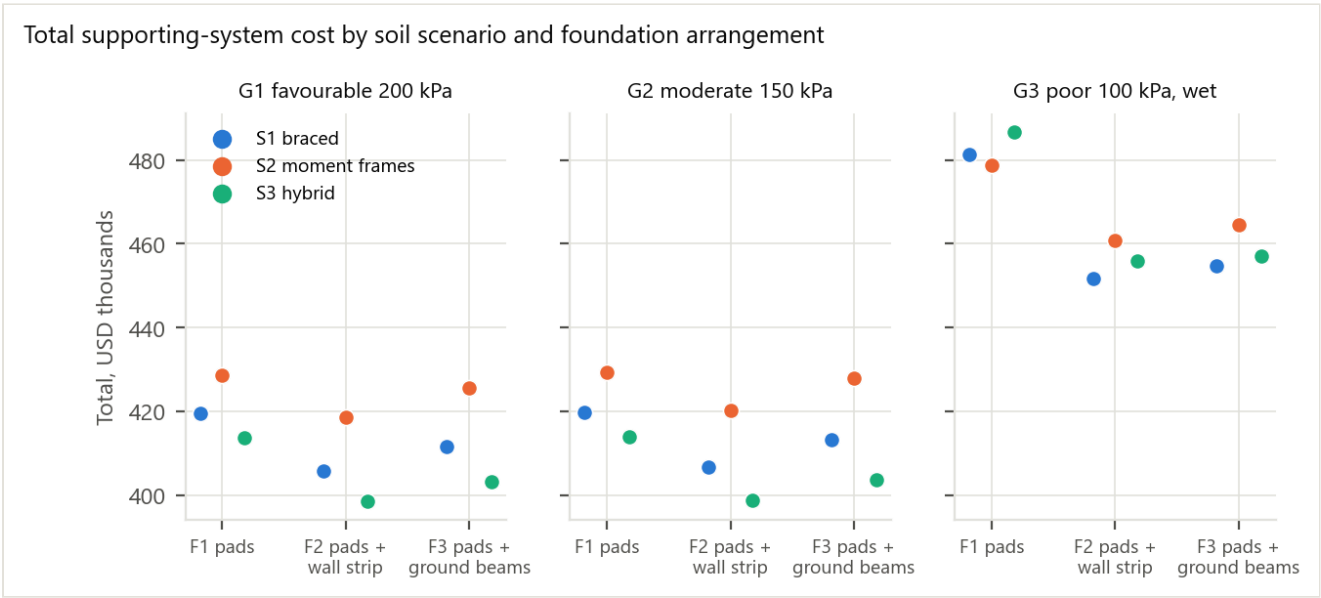


Figure 10.1 Total supporting-system cost by soil scenario and arrangement (each system at its selected column).

System	Soil	F1	F2	F3
S1 Braced frame	G1	419,574	405,780	411,607
S1 Braced frame	G2	419,651	406,705	413,339
S1 Braced frame	G3	481,318	451,760	454,717
S2 Moment-resisting	G1	428,713	418,591	425,632
S2 Moment-resisting	G2	429,295	420,305	428,025
S2 Moment-resisting	G3	478,661	460,756	464,562
S3 Hybrid	G1	413,662	398,590	403,328
S3 Hybrid	G2	413,947	398,792	403,634

System	Soil	F1	F2	F3
S3 Hybrid	G3	486,694	455,919	457,056

10.3 Governing actions and R2 concept pads

Uplift and overturning under the concurrent combinations govern, not bearing. In S3 the ordinary columns need pads of about 2.3–2.4 m square, the braced-bay columns 3.1–3.3 m, and the braced gable posts about 3 m, all 0.80 m deep. The depth is set by the anchorage: 450 mm embedment plus room to anchor the hairpins below the heads. R2's 2.0×2.0×1.0 m pinned pad suits ordinary S1 bearings. Its 3.5×3.5×0.70 m fixed-base pad is of the right order only for the braced bays. The wall ballast assumed for F2 is the weight actually present over 3.0 m at each pad, net of doors, the north loggia (no wall on the strip) and the east glazed slot. Its sensitivity to a lighter masonry (0.7 × 72 kN/m) is given in §15.

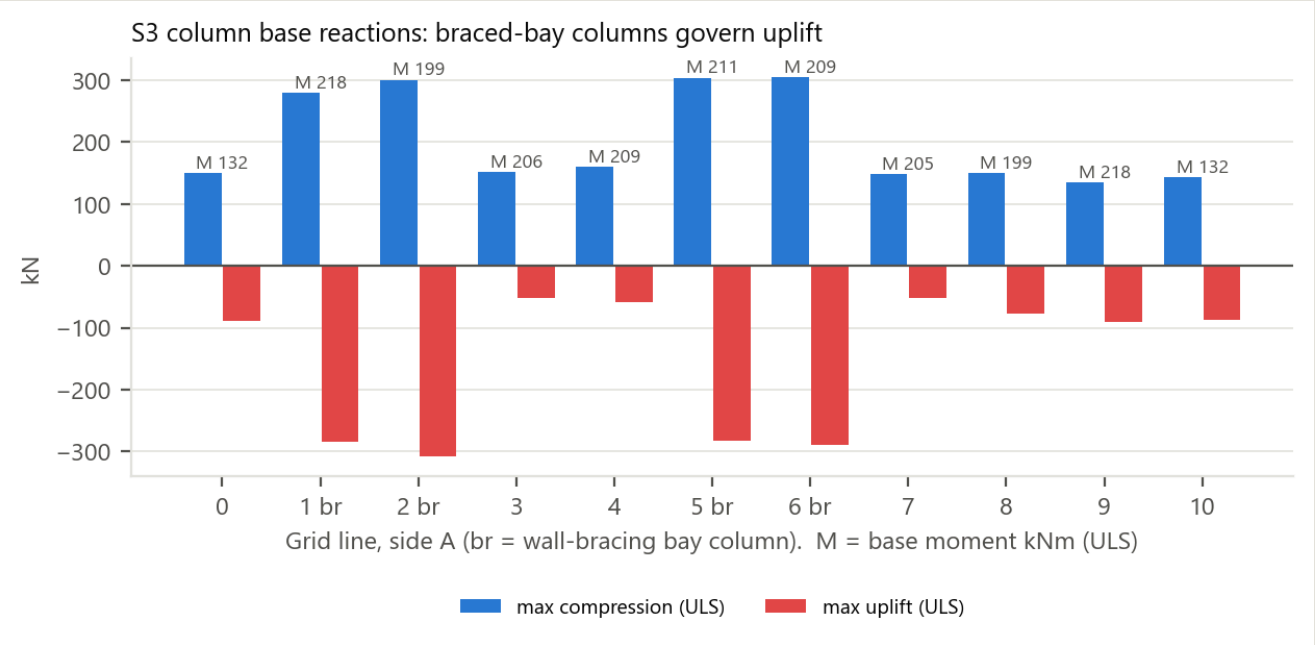


Figure 10.2 S3 base reactions along side A.

Base	N_c kN	N_t kN	H_y kN	H_x kN	M kNm	Base plate	Holding-down	Pad B×L×D m	A_s bot mm²/m	A_s top mm²/m
column 0A	150.2	89.7	45.3	19.8	131.9	608 x 455 x 30	4 M24 8.8 with anchor plates, hef 450 mm	2.3 x 2.30 x 0.80	1040	624
column 1A	280.1	285.1	74.4	57.2	218.3	608 x 455 x 30	8 M24 8.8 with anchor plates, hef 450 mm	3.3 x 3.30 x 0.80	1040	624
column 2A	300.5	308.6	63.9	58.6	199.3	608 x 455 x 30	6 M24 8.8 with anchor plates, hef 450 mm	3.1 x 3.10 x 0.80	1040	624
column 3A	151.2	52.7	64.4	0.0	205.7	608 x 455 x 30	6 M24 8.8 with anchor plates, hef 450 mm	2.4 x 2.40 x 0.80	1040	624
column 4A	160.4	58.7	64.7	0.0	209.2	608 x 455 x 30	6 M24 8.8 with anchor plates, hef 450 mm	2.4 x 2.40 x 0.80	1040	624
column 5A	303.8	282.6	64.8	57.8	211.0	608 x 455 x 30	6 M24 8.8 with anchor plates, hef 450 mm	3.1 x 3.10 x 0.80	1040	624
column 6A	304.4	289.3	64.7	56.6	209.0	608 x 455 x 30	8 M24 8.8 with anchor plates, hef 450 mm	3.2 x 3.20 x 0.80	1040	624
column 7A	148.6	51.6	64.4	0.0	205.3	608 x 455 x 30	6 M24 8.8 with anchor plates, hef 450 mm	2.4 x 2.40 x 0.80	1040	624
column 8A	149.5	77.1	63.9	0.0	198.6	608 x 455 x 30	6 M24 8.8 with anchor plates, hef 450 mm	2.4 x 2.40 x 0.80	1040	624
column 9A	134.5	90.7	74.4	0.0	218.2	608 x 455 x 30	8 M24 8.8 with anchor plates, hef 450 mm	2.5 x 2.50 x 0.80	1040	624
column 10A	142.9	87.7	45.3	19.8	132.1	608 x 455 x 30	4 M24 8.8 with anchor plates, hef 450 mm	2.3 x 2.30 x 0.80	1040	624

Base	N_c kN	N_t kN	H_y kN	H_x kN	M kNm	Base plate	Holding-down	Pad B×L×D m	A_s bot mm²/m	A_s top mm²/m
post 0 k1	15.9	-10.6	0.0	52.0	0.0	680 x 360 x 25	4 M24 8.8, hef 300 mm	1.1 x 1.65 x 0.60	780	0
post 0 k3	398.8	420.4	80.3	55.7	0.0	680 x 360 x 30	4 M24 8.8, hef 450 mm	3.3 x 3.30 x 0.80	1040	0
post 0 k5	467.9	363.1	70.7	61.8	0.0	680 x 360 x 30	4 M24 8.8, hef 450 mm	3.0 x 3.00 x 0.80	1040	0
post 0 k7	467.9	363.1	70.7	61.8	0.0	680 x 360 x 30	4 M24 8.8, hef 450 mm	3.0 x 3.00 x 0.80	1040	0
post 0 k9	398.8	420.4	80.3	55.7	0.0	680 x 360 x 30	4 M24 8.8, hef 450 mm	3.3 x 3.30 x 0.80	1040	0
post 0 k11	15.9	-10.6	0.0	52.0	0.0	680 x 360 x 25	4 M24 8.8, hef 300 mm	1.1 x 1.65 x 0.60	780	0

10.4 Integrated framing (gallery, cores, office)

Gallery: its slabs (+4.4 to +7.85 m) could restrain columns 7–9 at mid-height, but they do not reach the eaves, so they cannot shorten the girders. Using them would add lateral load to undesigned rakers and RC columns that have no footings. Not used. **Cores:** their walls stop at +10.80 m, 0.10 m below the truss bottom chord, and the core end walls (6.2 m long in x) cross under trusses 10 and 11. As designed RC shear walls with foundations they could take the east transverse and longitudinal reactions. That would remove the east gable bracing and the bay-5 wall bracing (≈3–4 t, USD 10–15k) but add core reinforcement and footings. Conditional. **Office wing:** its +4.42 m roof deck could prop the west gable posts, reducing the post size, but only if the office frame is braced for it. Not assumed.

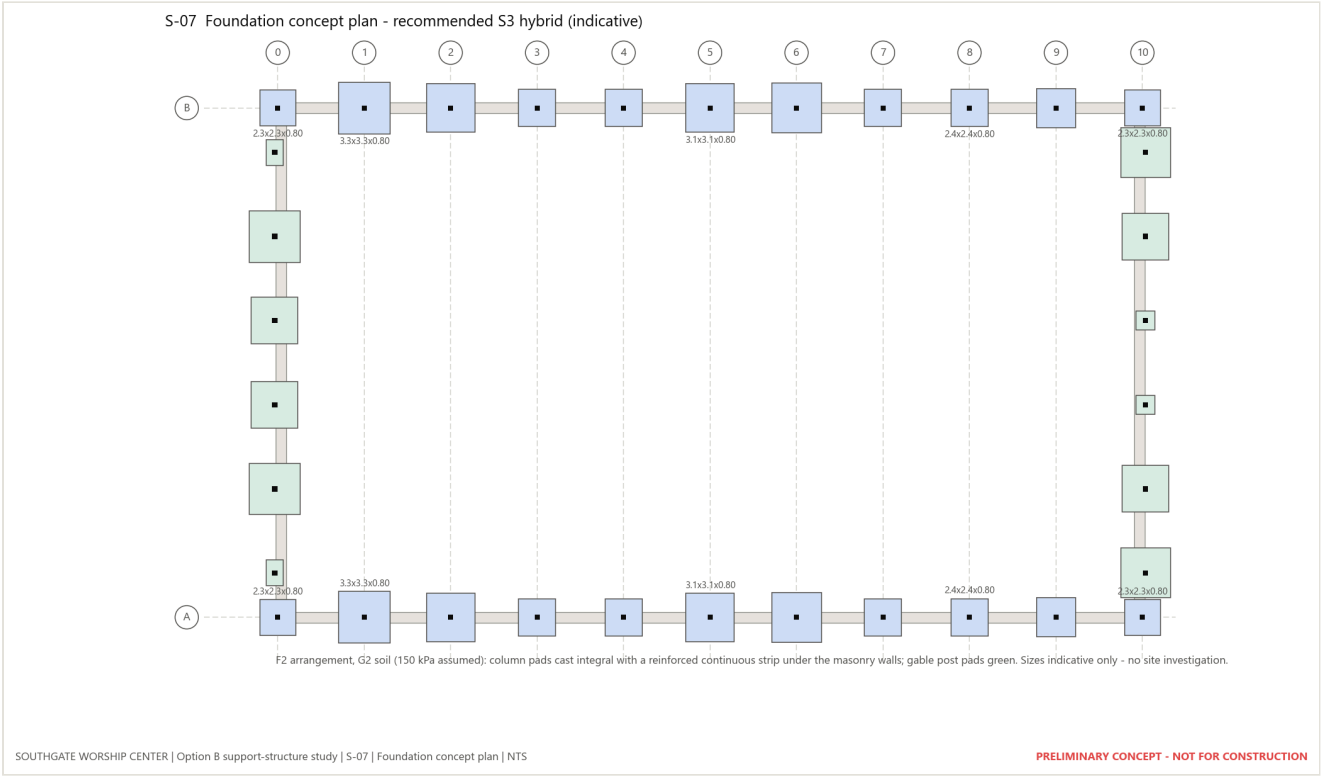


Figure 10.3 Foundation concept plan, recommended S3, F2, G2 (indicative).

11. Foundation reinforcement and anchorage considerations

- **Holding-down design uses concurrent actions.** The tension row is designed for the worst *concurrent* (N, M) pair from each ULS combination: $T = (M - N\,z/2)/z$ with compression, and $T = M/z + |N|/2$ with uplift. The independent review showed that combining the envelope maximum compression with the maximum moment under-estimated T by about 60% at the governing base. The S3 fixed bases need 6–8 M24 8.8 cast-in bolts with anchor plates (4 at the gable lines), hef 450 mm, 30 mm stiffened plates on 40–50 mm grout.
- **Concrete breakout:** without anchor reinforcement the cone utilisation exceeds 1 at most fixed bases and at the braced pinned bases. Designed supplementary reinforcement is therefore mandatory: hairpins around the tension bolts, anchored beyond the cone in both directions. That is why the pads are 0.80 m deep, and it is an inspection hold point. Four M24 bolts are not, by themselves, proof of anchorage.
- **Pads:** bottom mesh is the 0.13% minimum or more (≈1,040 mm^2/m at 0.80 m, e.g. Y16 @ 175). Top mesh is provided where uplift or base moment causes hogging. Punching includes moment transfer ($V_{eff} = V(1 + 1.5M/Vx)$, BS 8110 3.7.6.2), and one-way shear is checked to SANS 10100. The upward pull-out of the anchor group is resisted by the designed hairpins, not by plain concrete.
- **Wall strip (F2):** 0.60 × 0.45 m with ≈75 kg/m^3, continuous top and bottom bars lapped into the pads so that the engaged wall acts with each pad.
- **Shear transfer and sliding:** two mechanisms, priced separately. (i) Base plate to pad: friction 0.2N under the concurrent compression only; where any uplift combination carries shear, a stub shear key in a grout pocket is used. Bolts in shear through grout are not relied on. (ii) Pad to soil: where the concurrent horizontal force exceeds μN, a downstand key or passive resistance is provided (priced per pad).
- **Fixity:** the analysis assumes rigid base fixity. A rotational-spring sensitivity, $K = k_s LB^3/12$ halved for partial contact (≈50 MNm/rad for G2, ≈20 MNm/rad for G3), is reported in §15. The subgrade modulus must come from the site investigation.
- **Buoyancy:** in G3 a design water level of −1.00 m (0.40 m above the assumed perched level) is applied to all pads founded below it. The uplift resistance uses submerged densities.
- **Durability:** steel below FFL encased in C25/30. Holding-down assemblies hot-dip galvanised.

12. Structural performance comparison

Criterion	S1 Braced	S2 Moment	S3 Hybrid
Column	UC 305x305x137	UC 356x368x153	UC 305x305x97
Max column utilisation (ULS, masonry restraint 3.3 m)	0.526	0.346	0.609
Transverse column deflection SLS 0.85 kPa (limit 40.8 mm)	40.5	36.9	26.1
Column utilisation with no masonry restraint (13.08 m)	0.644	0.491	0.931
Column deflection at 1.20 kPa	57.1	52.2	36.8
Longitudinal eaves sway SLS 0.85 kPa (limit 40.8 mm)	31.8	25.9	40.7
Longitudinal eaves sway at 1.20 kPa (same design)	44.9	36.5	57.5
α_{cr} transverse / longitudinal	48.4 / 42.6	61.7 / 73.6	87.3 / 33.5
Max base moment kNm	0	322	218
Max column uplift kN	305	299	309
Gable-truss TC axial utilisation (collector)	≈0.92	≈0.65	≈0.84
Redundancy	single load path per direction	every line a frame (high)	girder + cantilever columns in parallel
Erection stability	needs temporary guys	self-stable	self-stable

All systems have $\alpha_{cr} > 10$, so first-order global analysis is adequate. Two member-level eigen-modes are not global and are not covered by this study's member checks. The gable bottom chord between ties ($\alpha \approx 2.0$) belongs to the Option B truss design and is reported to it. The braced gable posts about the minor axis (≈ 1.3–1.6) are why the posts get designed restraint struts. Longitudinal sway is at the limit for all systems by design (the wall bracing is stiffened until it passes). At 1.20 kPa every system needs heavier wall bracing; this is costed in the 1.20 kPa sensitivity.

13. Material quantity assessment

System	Column	Columns t	Posts t	Girders + bracing t	Plates t	Total steel t	Concrete m³	Rebar t	Excavation m³	HD bolts	Struct. bolts
S1 Braced frame	UC 305x305x137	39.4	15.1	24.6	5.2	**84.3**	190	6.1	649	136	1600
S2 Moment-resisting	UC 356x368x153	44.0	15.1	13.3	6.4	**78.8**	182	8.1	626	216	1380
S3 Hybrid	UC 305x305x97	27.9	15.1	24.8	5.8	**73.6**	229	8.9	746	184	1600

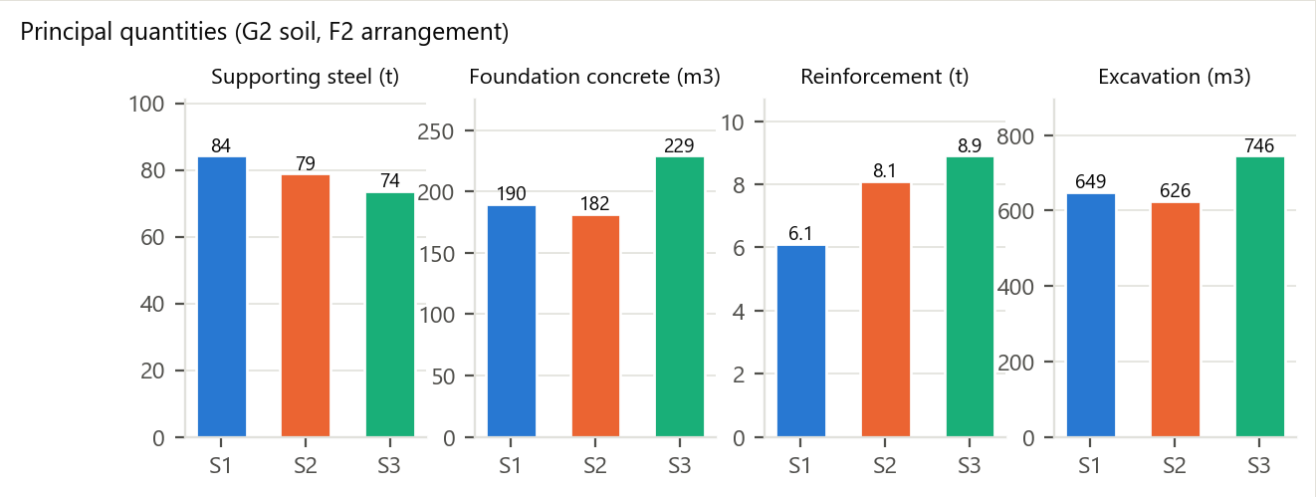


Figure 13.1 Principal quantities, G2/F2.

Basis: steel = member length × catalogue kg/m (calculated). Connections = 3 kg + 0.15 × kg/m per member end, plus bearing, base and post plates (allowance). Bolts = 2–4 per member end (preliminary). Footings, blinding, formwork, excavation (0.30 m working space, formation to FGL +0.10), backfill and disposal (bulking 1.25) = explicit geometry per pad (calculated from preliminary sizes). Reinforcement = designed A_s with 10% laps, plus 60 kg/m³ in encasement and 35–130 kg/m³ in the wall strip and ground beams (assumption-dependent). **Not reliably quantifiable yet:** bar schedules, masonry-tie restraints, transfer beam at the loggia, core and gallery foundations, dewatering. Quantities do not double-count: the wall strip is measured only between pads (pad lengths along the wall are deducted).

14. Construction cost comparison

USD, excl. VAT, Harare 2026, central rates (deliverable F gives low, central and high, with live formulas). The Option B trusses, purlins and equipment steel are **common** and excluded (≈USD 195k from R2, context only). Rates are attributable where possible (R2 Harare steel lists, Zimbabwe crane hire, a January 2026 Harare cement guide at USD 11–12.50 per 50 kg bag). Ready-mix, rebar per tonne, CHS and holding-down assemblies are **inferred**: no supplier quotation is included.

Cost component (USD)	S1 Braced	S2 Moment	S3 Hybrid
Side columns (supply)	66,813	91,517	44,614
Gable wind posts (supply, common)	25,680	25,680	25,680
Wind girders (supply)	8,364	0	7,365
Bracing, ties, struts (supply)	34,634	23,264	36,048
Connections, base plates, HD bolts, grout	20,212	24,344	23,192
Foundations: concrete, blinding, formwork	42,098	40,687	50,078
Reinforcement (pads, wall strip)	9,553	12,158	13,465
Groundworks: excavation, backfill, disposal	11,000	10,594	12,660
Fabrication and corrosion protection	58,936	60,119	58,836
Transport, erection, cranes, access, temporary works	64,191	55,687	59,713
QA/testing, O&P on labour and plant, contingency	56,724	71,255	62,141
Total supporting-system cost	**406,705**	**420,305**	**398,792**
Difference from S3	+7,913 (+2.0%)	+21,513 (+5.4%)	–

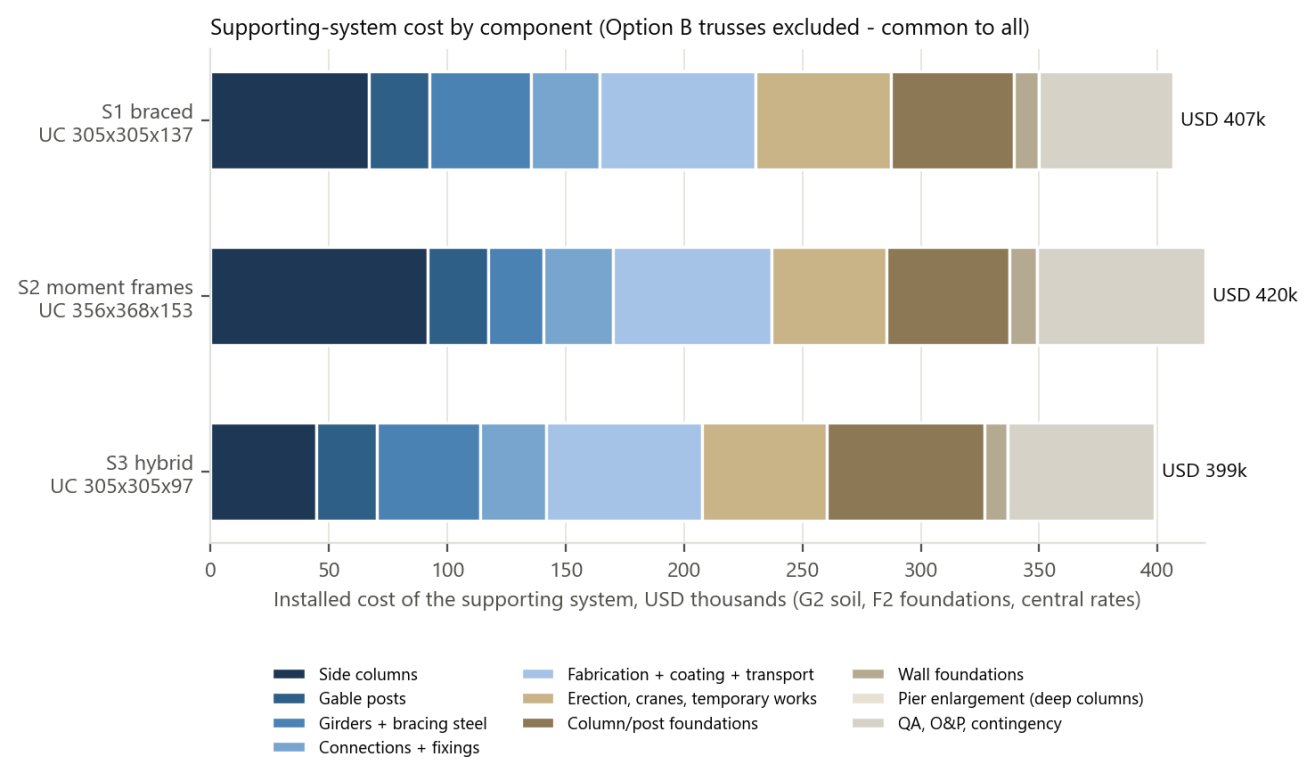


Figure 14.1 Cost by component.

S3's columns are 30% lighter than S1's and avoid S2's imported sections, but its fixed-base foundations cost ≈USD 10,927 more than S1's. The net difference (8k, 2.0%) is **smaller than the estimating uncertainty**. The headline figures are therefore the P50 values (§15), and the choice between S1 and S3 rests on ground conditions and non-cost factors. Contingency is 10% (design development, no SI) plus scheme risk: S2 +4% (imported sections, knee re-design), S3 +2% (fixed-base anchorage). Erection productivity: 6 columns, 4 posts and 15 bracing pieces per crane day, plus 2 days of girder assembly.

15. Geotechnical and cost sensitivity

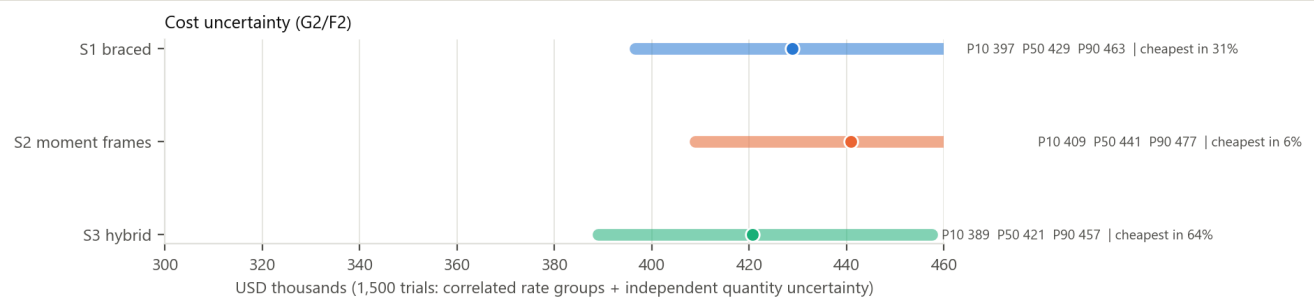


Figure 15.1 Monte Carlo: correlated rate groups and independent quantity uncertainty per scheme.

15.1 Structural sensitivity (each case fully re-sized)

Case	S1 UC 305x305x137	S2 UC 356x368x153	S3 UC 305x305x97
Base case	pass · u 0.53 · 40/32 mm	pass · u 0.35 · 37/26 mm	pass · u 0.61 · 26/41 mm
SLS wind at 1.20 kPa	FAIL · u 0.53 · 48/41 mm	FAIL · u 0.35 · 52/36 mm	FAIL · u 0.61 · 34/55 mm
Dominant opening cpi +0.55	FAIL · u 0.60 · 42/31 mm	pass · u 0.35 · 37/26 mm	pass · u 0.66 · 26/41 mm
Dominant opening in suction zone, cpi -1.08 (factored 1.3)	FAIL · u 0.53 · 55/31 mm	pass · u 0.45 · 37/28 mm	pass · u 0.70 · 37/40 mm
ULS wind 0.85 kPa only	pass · u 0.37 · 40/32 mm	pass · u 0.24 · 37/26 mm	pass · u 0.43 · 27/41 mm
Future rigging/equipment	pass · u 0.53 · 40/32 mm	pass · u 0.35 · 37/26 mm	pass · u 0.61 · 26/41 mm
Base fixity: rotational spring 50 MNm/rad (G2)	pass · u 0.53 · 40/32 mm	FAIL · u 0.34 · 55/26 mm	pass · u 0.62 · 30/41 mm
Base fixity: rotational spring 20 MNm/rad (G3)	pass · u 0.53 · 40/32 mm	FAIL · u 0.38 · 75/26 mm	pass · u 0.64 · 35/41 mm
Lightweight cladding, h/150	pass · u 0.61 · 48/37 mm (UC 305x305x118)	pass · u 0.42 · 44/29 mm (UC 356x368x129)	pass · u 0.61 · 28/43 mm

Cells: verdict · column utilisation · transverse / longitudinal SLS sway (mm). The bracing is re-sized in each case. The cpi −1.08 case is factored at 1.3, which is conservative if door management makes it an accidental situation (γ 1.0). Base-fixity cases apply to the fixed-base systems S2 and S3 only.

- **Monte Carlo:** correlated rate groups, plus independent quantity factors per scheme (steel 0.92–1.00–1.10 for the sizing method; foundations 0.85–1.00–1.25 for the missing SI). P50: S1 429k, S2 441k, S3 421k. P(S3 < S1) = 0.67; P(S3 < S2) = 0.89. **S3 against S1 is not decisive on cost alone.**
- **Soil (decision threshold):** cheapest system G1 S3, G2 S3, G3 S1. In G3 the buoyant, softer ground penalises the large fixed-base pads more than the pinned S1 pads, and S1 becomes cheapest. The SI therefore decides between S1 and S3. Below about 75 kPa allowable, or with high swell potential, a raft or piles would be needed (outside this study).
- **Masonry density 0.7 × 72 kN/m (hollow / lighter units):** S1: USD 407k; S2: USD 420k; S3: USD 399k (G2/F2). There is almost no effect, because the construction stage (walls absent) already governs most pads.
- **Wind, opening and fixity cases:** see the table above (each case re-sized for the selected column). **Structural robustness favours S3.** It is the only system that passes partial base fixity, both dominant-opening cases, and transverse SLS at 1.20 kPa. S1 fails both opening cases and transverse SLS at 1.20 kPa. S2 depends strongly on base fixity (55–75 mm with springs) and fails transverse SLS at 1.20 kPa.
- **SLS at 1.20 kPa, longitudinal:** all three systems exceed the limit longitudinally. The wall bracing in bays 1 and 5, which is common to all systems, would need a third braced bay per side or heavier members, an estimated +1.5–2.5 t and USD 6–10k for every system. This does not change the ranking. It must be resolved once the design wind is confirmed.

- **Decision threshold, wall construction:** with lightweight cladding (h/150) the masonry stiffness criterion no longer governs, and the lightest passing columns are those shown in the "Lightweight cladding" row. The wall type must be fixed before the stability system is frozen.

16. Architectural coordination

Element	Impact	Mitigation / status
UC 305×305×97 columns (regional stock, study judgement)	308–315 mm deep, within the 420 mm wall + pier zone at each grid; 8–15 mm beyond the 300 mm wall face if centred	Centre the column in the pier zone; adjust the pier cladding. No change to the footprint
Wall bracing bays 1 and 5	X below the clerestory plus knees in the piers. Visible on the inner face unless built into the masonry	Build in with compressible packing, or line internally. Clerestories, doors and circulation unaffected
Gable posts (UB 533)	533 mm deep, outside the truss plane; fit within the gable wall build-up only if the wall zone is ≥ 550 mm	West: behind the stage backdrop. East: in the stair cores. Wall thickening or exposed posts to agree
Gable bracing	West panels behind the stage screens (s 8.1–13.5, 18.9–24.3); east panels inside the stair cores	Avoids doors and the east glazed slot; screens to be coordinated
Eaves wind girders	In the roof plane within 5.4 m of each eave, above the ceiling/acoustic zone	Clear of catwalk zones; coordinate with the ridge fans (rows 11/13 unaffected)
Pads 2.5–3.6 m and wall strip	Below ground. Pads extend ≈1.2–1.7 m inside the wall line under the raked floor and external paving	Coordinate with drainage, the hall slab edge and services entries
Loggia transfer beam	Needed at +4.00 between lines 7–9 north	Architectural depth to agree (not in the model)
Glazed elevations, entrances, sightlines, gallery, office wing	No change	–

17. Recommended structural solution

Base case (G1/G2 ground, masonry walls): S3 Hybrid on F2 foundations (indicative member sizes, not for fabrication):

- 22 columns UC 305x305x97 S355, strong axis normal to the wall, fixed bases (stiffened 30 mm plate, 4–8 M24 8.8 with anchor plates, hef 450 mm, designed hairpins), propped at the eaves.
- Eaves wind girders both sides, TC plane, 5.4 m deep: diagonals CHS 168.3x5, inner chord CHS 139.7x4.5, eaves struts CHS 168.3x6.3 (purlins not relied on).
- End-bay roof girders CHS 168.3x5; cross frames CHS 139.7x4.5; bottom-chord bracing CHS 139.7x4.5; ties CHS 88.9x3.2 @ 5.4 m; heel restraint at the gable trusses.
- 12 gable posts UB 533x210x82 with two rows of restraint struts per gable; gable bracing CHS 193.7x6.3 (two-tier X, 2 panels per gable).
- Wall bracing bays 1 and 5: X CHS 219.1x8 below +9.90 (in two halves), knees CHS 114.3x5, sill CHS 101.6x3.6.
- Pads ≈2.3–3.3 m × 0.80 m C25/30 integral with a 0.60 × 0.45 m reinforced wall strip (G2). Re-size after the site investigation.
- Supporting steel ≈74 t; concrete ≈229 m^3; reinforcement ≈8.9 t; cost P50 ≈USD 421k (central 399k).

Alternative to adopt if the SI finds G3-type ground (or if fixed-base anchorage is unattractive to the contractor): S1 Braced, UC 305x305x137 on pinned 4 × M24 bases, the same roof girders and bracing, and smaller pads. P50 ≈USD 429k. S2 is not recommended in any scenario.

17.1 Approval gates

1. **G-A (before design freeze):** confirm masonry walls (if they are lightweight cladding, re-run); confirm the code set and the Harare design wind; agree the visibility of the wall bracing.
2. **G-B (before the stability system is fixed):** site investigation. Confirm q_a, groundwater, swell/collapse potential and subgrade modulus. G1/G2 → S3 + F2. G3 → S1 + F2/F3, or S3 + F3 if the swell potential requires ground beams anyway.
3. **G-C (before BIM implementation):** review of this study by the appointed ECZ engineer, then a controlled IFC update per deliverable J.

18. Outstanding engineering requirements

ID	Discipline	Issue	L	I	Action / information	Gate
R01	Architecture/Structure	Side-wall construction not confirmed: IFC elements are named 'BRICK-INFILL' (300 mm, source facade_system 'Brick masonry infill') but carry a metallic-cladding render material	High	High	Confirm wall construction (masonry type/thickness/density/reinforcement) before detailed design	G-A (before design freeze)
R02	Geotechnical	No site investigation exists; soil scenarios G1-G3 are assumptions	High	High	Trial pits (min. 6 to 3.0 m), 2 rotary/DCP boreholes or DPSH to 6 m, indicator tests, swell/collapse potential, groundwater observation, bearing recommendation	G-B (before foundation design)
R03	Wind	Harare design gust / peak velocity pressure not confirmed (0.85 and 1.20 kPa are R2 study assumptions)	Medium	High	Confirm code edition (SANS 10160-3:2019 or other) and Harare basic wind speed with Harare City Council and the appointed ECZ engineer	G-A
R04	Structure	Masonry restraint to column and post minor axes assumed at 3.3 m (ties/bond beams)	Medium	High	Adopt UC305x97; provide 2 rows of designed restraint struts per gable for the posts (included); specify wall ties as restraints only where designed	Detailed design
R05	Structure	Option B gable-truss top chord 2L80x8 used as lateral-system collector reaches axial utilisation ~0.84 (S3) / 0.92 (S1) before purlin bending	Medium	Medium	Include lateral-system actions in Option B detailed design; consider 2L90x8 gable top chord (as interior lines)	Detailed design
R06	Structure	Truss bearing at y = 0.30 m is 145 mm eccentric to the column axis (y = 0.155 m) as modelled	Medium	Low	Detail a concentric bearing (cap plate centred) or retain eccentric moment in design	Detailed design
R07	Structure	North wall masonry above the 9.5 m loggia opening (x 43-52.5 m) has no transfer beam in the model	High	Medium	Design a transfer beam at +4.00 between lines 7-9; confirm loads	Detailed design
R08	Structure	Gable-truss bottom-chord heel panels buckle under uplift without restraint (eigen alpha 0.22)	High (if omitted)	High	Provide out-of-plane heel restraint cleat at gable lines 0 and 10 (included in study model)	Detailed design
R09	Structure	Gable wind posts (UB533x210x82) and their bracing were not in the R2 scope; braced posts carry 330-560 kN ULS uplift	High	Medium	Optimise: widen gable bracing panels, prop west posts at the office roof (+4.42) if verified, or use designed masonry gable shear walls	Detailed design
R10	Structure	Longitudinal wall bracing must avoid clerestory windows; arranged as X below +9.90 with knee braces in 1.08 m piers (bays 1 and 5)	Medium	Medium	Agree bracing visibility/enclosure; detail masonry movement joints and soft packing around bracing	Design development
R11	Structure	Masonry infill tight to the steel frame will attract in-plane load before the steel bracing	Medium	Medium	Detail movement joints and flexible ties so the steel system governs (or design masonry as shear walls)	Detailed design
R12	Foundations	Construction-stage uplift before masonry is built (roof clad first)	Medium	Medium	Specify construction sequence or confirm temporary-condition wind with the engineer	Tender

ID	Discipline	Issue	L	I	Action / information	Gate
R13	Foundations	Fixed-base anchorage designed for concurrent N+M: 6-8 M24 groups exceed cone capacity without anchor reinforcement; pad depth 0.80 m set by anchorage	Medium	High	Design anchor (supplementary) reinforcement to EN 1992-4 / ACI 318 ch.17 or equivalent; site inspection hold point	Detailed design
R14	Foundations	Stair-core walls and gallery rear columns have no foundations modelled; core construction (RC or masonry) unverified	High	Medium	Design core and gallery foundations; decide whether cores join the lateral system (S3-core variant)	Design development
R15	Structure	East-gable columns (line 10) have no concept footing in the IFC (COL-0015/0016)	Certain	Low	Add pads in controlled BIM update (spec J)	BIM update
R16	Procurement	UC356x368 (S2) not rolled in SADC; import premium 20-45 % and lead time	Medium	Medium	Obtain supply quotations if S2 is pursued	Tender
R17	Procurement	Rates are inferred: no Harare quotation for UC305x97, CHS bracing, ready-mix, rebar or holding-down bolts	Medium	Medium	Tender enquiry to two or three Harare fabricators and concrete suppliers	Tender
R18	Code	Design standards not formally confirmed for Harare (SANS 10160 / 10162-1 / 10100-1 used by analogy; Zimbabwe Model Building By-laws)	Medium	Medium	Confirm the design-code set and editions with the authority	G-A
R19	Equipment	Rigging/equipment loads are scenarios (no AV/HVLS products selected)	Low	Low	Equipment schedule with weights and rated points	Design development
R20	Analysis	Study model is a linear 3D frame model verified against closed-form checks; not a commercial FE package; joint flexibility and bolt slip ignored	Medium	Medium	Independent 3D analysis in recognised software at detailed design; independent check	Detailed design
R21	Architecture	Fixed bases need larger pads (S3: ~2.5-3.6 m) under the wall line; interfaces with external paving and internal raked floor	Low	Low	Coordinate pad footprints with drainage and the hall floor slab edge	Design development
R22	Durability	Steel below FFL encased in concrete; corrosion category C2 assumed internally	Low	Medium	Confirm coating system, encasement and HD-bolt galvanising	Specification
R23	Foundations	Base fixity assumed rigid in design runs	Medium	Medium	Obtain subgrade modulus from SI; re-check fixity with rotational springs	G-B
R24	Geotechnical	Ground conditions decide between S3 (firm ground) and S1 (poor/wet ground)	High	Medium	Hold the stability-system decision until the SI (gate G-B)	G-B

19. Conclusions

1. The existing columns, bases, footings and bracing are concept geometry. They do not form a complete or adequate load path for the Option B roof and the masonry walls.
2. With 12.23 m of masonry spanning to the columns, the column criterion is stiffness (h/300 = 40.8 mm), in both directions, not strength. That moves the optimum away from R2's pinned UC 254 columns.
3. All three stability systems can be made to work. S2 (moment frames) is the most expensive in every scenario. S3 (hybrid) and S1 (braced) are within the estimating uncertainty of each other (2.0% at G2). S3 is preferred on firm ground for its lighter columns, erection self-stability and transverse stiffness reserve. S1 is preferred on poor or wet ground, where its smaller pinned-base pads are cheaper.
4. Foundations are governed by uplift and overturning under concurrent actions at the braced bays, the gable posts and the fixed bases. Anchorage, not bending, sets the pad depth. Integrating the pads with a reinforced wall strip is the cheapest arrangement in every case.
5. The ground and the wall construction are the assumptions that can change the recommendation. Both must be confirmed before the stability system is frozen.
6. The Option B truss remains appropriate. Its gable top chords, its bottom-chord heels and the bearing bolts need attention in its own detailed design.
7. The independent review changed the study materially: concurrent anchorage actions, purlin-independent girders, buoyancy, anchorage-led pad depth, a no-restraint robustness test, post restraint and slower erection raised all totals by about 20%, and turned a clear S3 preference into a conditional one (deliverable I).

20. Technical appendices

A. Solver verification (scripts/hand_checks.py)

Case	Model	Closed form	Ratio
cantilever tip deflection (mm)	320.92	320.92	1.0000
cantilever base moment (kNm)	130.00	130.00	1.0000
propped column mid deflection (mm)	83.04	83.51	0.9944
propped column max moment (kNm)	106.93	106.93	1.0000
fixed-base propped column max deflection (mm)	34.70	34.67	1.0009
fixed-base propped column base moment (kNm)	106.74	106.93	0.9983
Euler load pinned column (kN)	1316.45	1316.44	1.0000
two-bar truss member force (kN, compression)	70.71	70.71	1.0000

B. Independent hand checks of key results

Quantity	Hand calculation	Model	Agreement
UC 254×73 pinned column deflection, q 0.85	$5wL^4/384EI$, w = 0.85×1.02×5.56 = 4.82 kN/m, L 13.08 → 80.5 mm	81 mm column + ≈20 mm girder = 101 mm	consistent
UC 305×97 fixed-base propped deflection	$wL^4/185EI$ = 17.1 mm, + girder ≈11 mm	28.7 mm	consistent
S3 base moment, ULS 1.20	$wL^2/8$, w = 1.3×1.2×1.02×5.56 = 8.85 kN/m → 189 kNm (+ zone A, eccentricity)	198–218 kNm	consistent
Eaves girder (S1), both walls shared	net 1.09 q_p; ULS 5.6 kN/m per girder; M = 2,150 kNm; chord 398 kN (simple span, columns not sharing)	R2 2,135 kNm / 395 kN; design model with purlins as nominal links: see D (columns and gable bracing share part of the load)	consistent
Bearing uplift, ULS 1.20, near-edge line ($c_{p,net}$ −0.8)	(−0.8×1.2×1.3 + 0.9×0.36) × 90 m^2 ≈ −83 kN; plus windward-ridge zones in transverse wind	−92 kN (lines 1–2)	consistent within 10%
Braced-bay uplift	H ≈ 200 kN × 13.1 m / 5.56 m ≈ 470 kN gross, less gravity	285–313 kN	order of magnitude

C. Model limitations

- Linear elastic 3D frame, small displacements. Webs and bracing pin-ended. Joint flexibility, bolt slip and gusset eccentricities ignored. Masonry not modelled structurally (loads only; minor-axis restraint assumed in the checks).
- Member checks are in the form of SANS 10162-1, with simplified LTB and the β-factor. Purlins are modelled as axial struts only. Truss member design is not re-checked (R2's scope), except for the collector forces reported.
- Second order by $\alpha_{cr} \geq 10$ (no amplification required). The eigen-analysis is used diagnostically.
- Results are reliable for comparing and ranking systems and for preliminary sizing. They are not a compliance check. An independent analysis in recognised software is required at detailed design.

D. Sources

- Roof Structure Options Appraisal R2 and its data (`F-site-A-2000-astra-v2/studies/roof-truss-selection`, read-only).
- Southgate BIM audit reports: bim_quality_audit.md, migration_completeness.md, independent_final_review.md (O-04, O-06), review_response.md, outstanding_work.csv.
- Harare building materials guide (cement USD 11–12.50/bag), urbanedgezw.blogspot.com, Jan 2026. UZ structural cracks case study (ir.uz.ac.zw/handle/10646/3566). Zimbabwe soil classification (CCARDESA). Trade-barriers listing of SAZ concrete code / SI 41 of 1978. Herald: Model Building By-laws.
- SANS 10160 parts 1–3, SANS 10162-1:2011, SANS 10100-1, EN 1991-1-4, EN 1992-4, EN 1993-1-1 (by reference; not reproduced).

E. Independent review

A separate AI reviewer, given the code and outputs but not the author's reasoning, raised 16 objections (2 critical, 8 major). All were addressed by changing the calculations, not the text; the record is deliverable I (`review/I_independent_review_record.md`). This is an AI-assisted internal challenge, **not** independent professional engineering certification.